

3.3.6 Hypothesis Testing and Statistical Significance Framework

To determine whether observed differences in risk-adjusted performance (Sharpe and Sortino ratios) and weekly portfolio returns between candidate strategies and baseline benchmarks are statistically meaningful or merely artifacts of sampling variation, this study implements a decision-theoretic inferential hypothesis testing framework. In emerging equity markets such as the NGX—where asset returns display pronounced non-normality, negative skewness, and heavy tails—traditional parametric Z-tests can yield misleading p-values. Therefore, both parametric and robust non-parametric bootstrap procedures are employed:

1. Jobson and Korkie (1981) Test with Memmel (2007) Correction:

Evaluates the null hypothesis of Sharpe ratio equality $H_0: \text{Sharpe}_A = \text{Sharpe}_B$ for two correlated portfolios. Memmel (2007) corrects the asymptotic variance under portfolio correlation ρ :

$$Z = (\text{Sharpe}_A - \text{Sharpe}_B) / \sqrt{(1/T) [2(1 - \rho) + 0.5(\text{Sharpe}_A^2 + \text{Sharpe}_B^2 - 2 \text{Sharpe}_A \text{Sharpe}_B \rho^2)]}$$

where T is the number of out-of-sample weekly observations.

2. Ledoit and Wolf (2008) Circular Block Bootstrap Test for Sharpe Ratio Equality:

Resamples overlapping blocks of length $b = 5$ weeks across $B = 2,000$ bootstrap replicates to preserve time-series autocorrelation and conditional heteroskedasticity (GARCH effects). The empirical p-value evaluates $H_0: \text{Sharpe}_A - \text{Sharpe}_B = 0$.

3. Ledoit and Wolf (2011) Bootstrap Test for Sortino Ratio Equality:

Extends non-parametric block bootstrapping to downside risk, evaluating $H_0: \text{Sortino}_A - \text{Sortino}_B = 0$ to verify excess return per unit of downside risk.

4. Non-Parametric Wilcoxon Signed-Rank Test and Paired t-Test:

Evaluates whether weekly differential returns $\Delta R_t = R_{A,t} - R_{B,t}$ significantly deviate from zero ($H_0: E[\Delta R_t] = 0$).

5. Diebold and Mariano (1995) Test for Predictive Accuracy:

Evaluates whether out-of-sample forecasting loss (RMSE) differentials between machine learning models (Random Forest, XGBoost) and the Historical Mean baseline are statistically significant.